Model Question Paper-1 with effect from 2018-19 (CBCS Scheme)

USN | | | | | | | | | | |

Fifth Semester B.E. Degree Examination

Database Management System

TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
02.
03.

		Module – 1	**Marks**
Q.1	**(a)**	List and briefly explain the characteristics of Database Approach	8
	(b)	Briefly explain the Following i.Database ii. DBMS	2
	(c)	List and briefly explain the advantages of using DBMS Approach	10
		OR	
Q.2	**(a)**	Briefly explain the following i. Database Schema ii. Weak Entity Type	2
	(b)	With a neat diagram, explain the Three Schema Architecture	8
	(c)	Draw an ER Diagram for University Database by considering at least 5 entities	10
		Module – 2	**Marks**
Q.3	**(a)**	List & briefly explain the characteristics of Relations	8
	(b)	Briefly explain the following i. Domain Constraints ii. Referential Integrity Constraints	2
	(c)	By refereeing the following Database schema. Employee(Fname, Minit, Lname, <u>SSN</u>, Bdate, Address, Sex, Salary, Sup_SSN,Dno) Department(Dname, <u>Dnumber</u>, Mgr_SSN, Mgr_Start_date) Dept_Locations(<u>Dnumber</u>, <u>Dlocation</u>) Project(Pname, <u>Pnumber</u>, Plocation, Dnum) Works_On(<u>Essn</u>, <u>Pno</u>, Hours) Dependent (<u>Essn</u>, <u>Dependent_Name</u>, Sex, Bdate, Relationship) **Write the relational algebra expressions for the following queries** (i) Retrieve all the employee names who are working for department number 5. (ii) Retrieve all the projects which are controlled by department number 4. (iii) Retrieve the names of employees who have no dependents. (iv) Retrieve all the Employee Name who is working on all the projects in which John Smith works on. (v) Retrieve all the project numbers along with number of employee working on each project	10
		OR	
Q.4	**(a)**	With an example explain the steps of ER to Relational Mapping Algorithm	10
	(b)	List and briefly explain the various attribute Data Types and Domains in SQL	7

	(c)	With an explain, UPDATE Statement in SQL	3
		Module – 3	**Marks**
Q.5	**(a)**	By refereeing the following Database schema. Employee(Fname, Minit, Lname, SSN, Bdate, Address, Sex, Salary, Sup_SSN,Dno) Department(Dname, Dnumber, Mgr_SSN, Mgr_Start_date) Dept_Locations(Dnumber, Dlocation) Project(Pname, Pnumber, Plocation, Dnum) Works_On(Essn, Pno, Hours) Dependent (Essn, Dependent_Name, Sex, Bdate, Relationship) **Write the SQL Queries for the following** (i). Retrieve the name and address of all employees who work for the 'Research' department. (ii). Make a list of all project numbers for projects that involve an employee whose last name is 'Smith', either as a worker or as a manager of the department that controls the project. (iii). List the names of managers who have at least one dependent. (iv). Find the sum of the salaries of all employees, the maximum salary, the minimum salary, and the average salary. (v). For each project, retrieve the project number, the project name, and the number of employees who work on that project.	10
	(b)	With an example, explain Specifying Constraints as Assertions in SQL	5
	(c)	With an example, explain the concept of View in SQL	5
		OR	
Q.6	**(a)**	With an example, explain Cursors in Embedded SQL.	5
	(b)	List & briefly explain different types of JDBC Drivers	5
	(c)	With a neat diagram, Explain Three Tier Architecture Database Applications and briefly explain the advantages of Three tier Architecture	10
		Module – 4	**Marks**
Q.7	**(a)**	With an suitable explain Informal Design Guidelines for Relation Schemas	10
	(b)	With an suitable explain, First Normal Form, Second Normal Form, & Third Normal Form	10
		OR	
Q.8	**(a)**	With an suitable example, explain Properties of Relational Decompositions	10
	(b)	Consider the relation REFRIG(Model#, Year, Price, Manuf_plant, Color), which is abbreviated as REFRIG(M, Y, P, MP, C), and the following set *F* of functional dependencies: $F = \{M \rightarrow MP, \{M, Y\} \rightarrow P, MP \rightarrow C\}$ (i) Evaluate each of the following as a candidate key for REFRIG, giving reasons why it can or cannot be a key: {M}, {M, Y}, {M, C}. (ii) Based on the above key determination, state whether the relation REFRIG is in 3NF and in BCNF, giving proper reasons. (iii) Consider the decomposition of REFRIG into $D = \{R1(M, Y, P), R2(M, MP,C)\}$. Is this decomposition lossless?	10
		Module – 5	**Marks**
Q.9	**(a)**	List and explain ACID Properties	5
	(b)	Briefly explain two phase locking protocol	5
	(c)	Consider the three transactions *T*1, *T*2, and *T*3, and the schedules *S*1 and *S*2 given below. Draw the serializability (precedence) graphs for *S*1 and *S*2, and state whether	10

		each schedule is serializable or not. If a schedule is serializable, write down the equivalent serial schedule(s). $T1$: $r1$ (X); $r1$ (Z); $w1$ (X); $T2$: $r2$ (Z); $r2$ (Y); $w2$ (Z); $w2$ (Y); $T3$: $r3$ (X); $r3$ (Y); $w3$ (Y); $S1$: $r1$ (X); $r2$ (Z); $r1$ (Z); $r3$ (X); $r3$ (Y); $w1$ (X); $w3$ (Y); $r2$ (Y); $w2$ (Z); $w2$ (Y); $S2$: $r1$ (X); $r2$ (Z); $r3$ (X); $r1$ (Z); $r2$ (Y); $r3$ (Y); $w1$ (X); $w2$ (Z); $w3$ (Y); $w2$ (Y);	
		OR	
Q.10	**(a)**	With an example, explain basic Timestamp Ordering algorithm for Concurrency Control	6
	(b)	Briefly explain Transaction Support in SQL	4
	(b)	With an example, explain ARIES Recovery Algorithm	10